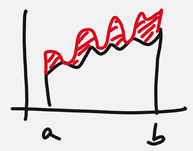
b- Si en $[a,b]$, $f \le g$, entonces, $\int_a^b f \le \int_a^b g$



$\int_a^{+\infty} g$ cv $\Rightarrow$ G es acotada $\Rightarrow$ F es acotada $\Rightarrow$ $\int_a^{\infty} f$ cv
(y acotada) (y creciente)

b-ii- $\int_a^x f \xrightarrow[x\to+\infty]{} +\infty$, $\int_a^x f \le \int_a^x g$

$\forall \Pi > 0, \exists x_0 \mid x \ge x_0 \Rightarrow F(x) \ge \Pi$.

$\forall \Pi > 0, \exists x_0 \mid x \ge x_0 \Rightarrow G(x) \ge \Pi$.

$G(x) \ge F(x) \ge \Pi$

↳ $\int_a^x g \xrightarrow[x\to+\infty]{} +\infty$.

Ex 2 **a-** $\int_2^{+\infty} \frac{1}{x \log^{\alpha} x}\,dx$ $\left(= -\int_0^{\ln(2)} \frac{1}{u \log^{\alpha} u}\,du\right)$.

$u = 1/x$: $dx = d(1/u)$

$x = 2 \Rightarrow u = 1/2$, $x \to +\infty \Rightarrow u \xrightarrow[u>0]{} 0$

$$\frac{dx}{x \log^{\alpha} x} = \frac{-du}{u \log^{\alpha}(1/u)} = \frac{du}{u \log^{\alpha} u}$$

$u = \ln x$, $du = \frac{1}{x}dx$, $x = 2 \to u = \ln 2$

$$I = \int_{\ln 2}^{+\infty} \frac{e^u\,du}{e^u (u)^{\alpha}} = \int_{\ln(2)}^{\infty} \frac{du}{u^{\alpha}} \quad (x \to +\infty,\ u \to +\infty)$$

$\int u^{\alpha} = \frac{1}{1-\alpha} u^{-\alpha+1}$ (+c)

Si $\alpha \neq 1$ $\lim_{x\to+\infty} \frac{u^{-\alpha+1}}{-\alpha+1} - \frac{\ln(2)^{-\alpha+1}}{-\alpha+1}$ c

$$= \begin{cases} +\infty & \text{si } -\alpha+1 \ge 0 \\ 0 - c & \text{si } -\alpha+1 < 0 \end{cases} \longrightarrow \text{DV si } \alpha \le 1$$

cv si $\alpha > 1$

Si $\alpha = 1$ $\int_{\ln(2)}^{\infty} \frac{1}{u}\,du = \lim_{u\to\infty} \ln(u) - \ln(\ln(2)) = +\infty$

$\int_2^{\infty} 1/x$ $\left(\sim \sum 1/n\right)$ DV

$$\frac{1}{x \log^{\alpha} x} = \frac{\log^{-\alpha} x}{x}$$

b- $\int$